



Environmental and herd management factors affecting the mineral profile of bovine milk

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ABSTRACT

This study investigated how animal, environmental, and management factors influence milk mineral composition (Ca, P, Mg, S, K, Cl, and Na) of Brown Swiss cows. A total of 1,060 individual milk samples were collected from 53 herds located in the Emilia-Romagna, Lombardia, and Toscana regions of Italy between February 2021 and April 2022, representing a wide range of production environments and management systems. Information on DIM, parity, and milk yield (kg/d) was combined with herd-level data on altitude, season, temperature–humidity index (THI), ventilation, and feeding system. Milk Ca, P, Mg, S, K, Na, and Cl (mg/kg) were determined using wavelength-dispersive X-ray fluorescence. A linear mixed model was used to estimate the effects of fixed animal-related, environmental, and herd management factors while accounting for herd as a random effect. Season significantly affected Ca, P, Mg, S, and K; THI was associated with Mg and Cl, particularly beyond the thermal stress threshold; altitude showed a tendency to be associated with K concentration; ventilation showed tendency of association with P and K, whereas feeding system was associated with P and S. These results demonstrate that, although the animal factors exerted the strongest influence, environmental and herd management factors can still modulate mineral variability. These findings refer to the studied Brown Swiss population and indicate that accounting for these factors is relevant to optimize milk quality, enhance cheese-making efficiency, and support precision management strategies in Italian dairy systems.

Key words: milk mineral, season, THI, ventilation, feeding system

INTRODUCTION

In recent decades, progress in genetics and herd management has led to remarkable gains in milk yield across modern dairy systems. However, despite an increasing emphasis on milk quality, attention has largely focused on fat and protein, leaving the mineral composition of milk comparatively underexplored. Minerals are fundamental not only for cheese-making and processing performance but also for the nutritional value of dairy products. However, the extent to which environmental conditions, management practices, and animal physiology influence mineral concentrations in milk remains insufficiently understood. Milk mineral concentration depends on multiple and interacting drivers. Scientific literature consistently demonstrates that macrominerals such as calcium (Ca), magnesium (Mg), potassium (K), sodium (Na), and phosphorus (P), as well as several microminerals, exhibit measurable genetic variability and moderate to high heritability. This evidence indicates that genetic background plays a role in determining milk mineral composition (Hossein-Zadeh, 2022; Singh et al., 2024).

As regards the effects of environment and management on milk mineral composition, most of the results are dated and may not fully reflect current dairy production systems or feeding practices. For example, the markedly higher productivity of modern dairy cows has been accompanied by altered metabolic and physiological demands (Martens, 2023), which in turn may have modified mineral utilization and secretion pathways. Moreover, advances in climate-mitigation strategies (e.g., enhanced ventilation and cooling) have significantly redesigned the dairy production environment (Sammad et al., 2020), underscoring the need to study milk mineral contents under current conditions.

Another key aspect concerns the analytical technique used for mineral quantification, as the choice of the analytical approach strongly affects detection limits,

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

precision and accuracy, and matrix recovery, making comparison across studies difficult (Soares et al., 2023). Several published results on milk minerals are from indirect estimates or predictive models rather than direct quantification (Manuelian et al., 2018; Visentin et al., 2019; Liu et al., 2025). This is largely due to the high cost and logistical complexity associated with mineral determination, particularly when large numbers of samples are involved. Although spectroscopic or mid-infrared prediction models provide rapid and cost-effective alternatives for large data sets, their calibration and validation depend on the quality of the reference data. For this reason, wavelength-dispersive X-ray fluorescence (WDXRF) could be of particular interest, as it enables direct quantification of all relevant elements, including Cl and S, in liquid milk without prior acid digestion. Unlike plasma-based techniques, which typically require mineralization of the organic matrix before analysis, WDXRF reduces sample preparation steps and allows multi-element determination in a single analysis, making it particularly suitable for comprehensive mineral profiling at population level. Nonetheless, when working on the genetic front, large-scale approaches remain necessary. Furthermore, research exploring environmental influences remains limited, as detailed herd-level data (e.g., feeding strategies, housing systems) are often difficult to access or systematically collect.

Although the influence of stage of lactation or breed on milk minerals has been proven to be relevant (Petrera et al., 2016; Manuelian et al., 2018; Stocco et al., 2019b), the lack of recent large-scale studies specifically addressing animal- and environment-related factors limits the applicability of earlier findings to current high-producing herds. These gaps highlight the need for updated field-based research that accounts for the complexity and variability introduced by modern genetics, environmental challenges such as heat stress and altitude, and evolving management systems. Accordingly, the aim of this study was to investigate the variability of milk minerals (Ca, P, Mg, S, K, Cl, and Na) in individual bovine milk samples by evaluating animal (days in milk, parity, and milk yield), environmental (altitude, season, and temperature–humidity index [THI]), and herd management factors (ventilation and feeding system).

MATERIALS AND METHODS

The cows involved in this study were not subjected to any invasive procedures, and milk samples were collected noninvasively by trained farm technicians as part of routine milking practices. No additional handling or intervention beyond standard herd operations was required, and thus approval by an Institutional Animal Care

and Use Committee or Institutional Review Board was not required.

Experimental Design

Individual milk samples were collected from 1,060 Brown Swiss cows (1 sample/cow) during the evening milking to allow analyses to be performed within 24 h of collection. Cows were reared in 53 commercial herds located across the Emilia-Romagna, Lombardia, and Toscana regions of Italy. Herd visits were systematically conducted from February 2021 to April 2022 (1 visit/herd), and DIM, parity, and milk yield (MY; kg/d) were recorded. From each herd, 20 cows were sampled and a milk aliquot of 50 mL per cow was used for the determination of its composition and mineral profile. Milk samples (collected without preservatives) were immediately cooled to 4°C and transported to the MilCA Laboratory of the University of Parma (Parma, Italy) for determination of milk gross composition, and to the RumiLab of Nutristar S.p.A. (Reggio Emilia, Italy) for mineral quantification. All samples were analyzed within 24 h of collection.

Environmental and Herd Characteristics

Each herd was characterized through a structured questionnaire designed to obtain comprehensive data on infrastructure, management practices, and milking procedures. This survey provided detailed information enabling the evaluation of environmental variables and the development of well-defined classification factors. Collected data included environmental factors (altitude, and THI on milk sampling day), as well as housing characteristics (i.e., bedding material, and access to outdoor paddocks), mechanical ventilation, milking system, and feeding strategy. Candidate variables were defined a priori based on biological plausibility. Model refinement was performed using stepwise selection based on Akaike's information criterion implemented in R (function *step*, package *stats*; R Core Team, 2024) to ensure parsimony and limit multicollinearity. The final model retained altitude, season, THI, ventilation, and feeding system (Table 1), together with the established animal factors.

Regarding the environmental factors, the altitude of each herd location was classified into 3 categories: (1) lowland or plains areas, including herds located within 300 m above sea level (a.s.l.; $n = 37$); (2) hilly areas, encompassing herds located from 300 to 600 m a.s.l. ($n = 10$); and (3) mountainous areas, comprising herds located above 600 m a.s.l. ($n = 6$). Seasons were defined according to calendar months: spring, summer, autumn,

Table 1. Distribution of sampled herds and individual cows across altitude, season, THI class, ventilation, and feeding system

Variable	Herds, n	Cows, n	Frequency, ¹ %
Altitude			
Plain	37	740	70
Hill	10	200	19
Mountain	6	120	11
Season			
Spring	18	360	34
Summer	11	220	21
Autumn	16	320	30
Winter	8	160	15
THI class			
≤67	35	700	66
68–72	5	100	9
73–76	7	140	13
≥77	6	120	11
Ventilation			
Yes	45	900	85
No	8	160	15
Feeding system			
Dry TMR	27	540	51
Traditional	17	340	32
Silage TMR	9	180	17

¹Frequencies (in %) are applied to both herds and individual cows, because the same number of cows (n = 20) was sampled from each herd.

and winter, corresponding to standard meteorological divisions.

The THI was calculated using temperature and humidity data of the geographic location and sampling date of each herd, using the formula proposed by Mader et al. (2006):

$$\text{THI} = (0.8 \times \text{Tdb}) + [(\text{RH}/100) \times (\text{Tdb} - 14.4)] + 46.4,$$

where Tdb is dry-bulb air temperature (°C) and RH is relative humidity (%).

Climatic information was retrieved from the NASA Prediction of Worldwide Energy Resource (POWER) Data Access Viewer (Sparks, 2018) via the *nasapower* R package (Blain et al., 2024). For each herd, latitude, longitude, and sampling date were used to extract daily average values of air temperature (°C) and relative humidity (%) at 2 m above ground for the 7 d preceding sampling. The maximum THI value recorded during this period was selected for each herd and categorized into 4 classes: ≤67 (n = 35), 68 to 72 (n = 5), 73 to 76 (n = 7), and ≥77 (n = 6). The same THI value was assigned to all cows sampled during the herd visit. Therefore, THI variation occurred only between herds and not within herd across seasons. Use of the maximum THI over the week preceding sampling was intended to capture the cumulative and delayed biological effects of heat stress, as milk production and mammary gland responses are known to occur with a lag of several days and to be more sensitive to extreme thermal conditions than to daily mean values (Collier et

al., 1981; West, 2003; Wildridge et al., 2018). Moreover, the use of a short presampling window reduces sensitivity to transient meteorological perturbations, which are particularly frequent in hilly and mountainous areas.

In addition, to partially account for potential discrepancies between outdoor climatic conditions and in-barn microclimate, herd-level mechanical ventilation (presence vs. absence) was included in the statistical model as a fixed effect.

With respect to diet, herds were classified into 3 feeding systems: dry TMR (**dTMR**), silage TMR (**sTMR**), and traditional system (**TRD**). The dTMR was based primarily on dried forages, including various legumes and grasses, combined with maize and soybean meal to provide a balanced intake of starch and protein. The sTMR relied on ensiled forages such as maize, wheat, alfalfa, or mixed silages, combined with ground grains and concentrates to ensure adequate nutrient supply. The TRD reflected long-established feeding practices that remain common in smaller herds or those located in mountainous areas. Under this system, cows were typically fed dried forages of different types, with concentrates administered separately, most often during milking.

Milk Composition and Mineral Profile

Milk fat, protein, casein, lactose, and total solids (%) were determined through an infrared spectrometer (Milkoscan FT3, Foss Electric, Hillerød, Denmark), calibrated according to reference methods ISO 12966-2 for fat (ISO, 2017), ISO 8968-1 | IDF 20-1 for protein (ISO and IDF, 2014), and ISO 26462 | IDF 214 for lactose (ISO and IDF, 2010).

Milk mineral concentration (mg/kg) was measured using WDXRF (S6 Jaguar, Bruker Corporation, Billerica, MA). To ensure measurement accuracy, the WDXRF instrument was calibrated using inductively coupled plasma optical emission spectroscopy for Ca, P, Na, K, Mg, and S, and ion chromatography for Cl. An innovative technique applied to mineral analysis in milk, WDXRF spectroscopy allows assessment of the mineral composition of a sample by exploiting the fluorescence radiation from X-rays. This method is based on the photoelectric effect, where incident X-ray radiation interacts with the atoms of the sample, leading to excitation of electrons and subsequent emission of fluorescence radiation at longer wavelengths. The unique fluorescence radiation emitted by each chemical element enables quantification of constituent atoms within the sample. In the case of liquid samples such as milk, ~3 mL/cow was introduced into a disposable cup and loaded into the instrument without any sample preparation. During analysis, the reading chamber is saturated with helium to optimize

measurement accuracy. The WDXRF instrument is capable of processing up to 20 samples, with an average analysis time of ~7 min per sample.

Statistical Analysis

Milk mineral concentration was analyzed using the following linear mixed model:

$$y_{ijklmnopqr} = \mu + \text{DIM}_i + \text{parity}_j + \text{MY}_k + \text{altitude}_{l(q)} \\ + \text{season}_{m(q)} + \text{THI}_{n(q)} + \text{ventilation}_{o(q)} \\ + \text{feeding system}_{p(q)} + \text{herd}_q + e_{ijklmnopqr},$$

where $y_{ijklmnopqr}$ is the mineral concentration (Ca, P, Mg, S, K, Cl, or Na); μ is the overall intercept of the model; DIM_i is the fixed effect of the i th class of days in milk ($i = <60, 61\text{--}120, 121\text{--}180, 181\text{--}240, 241\text{--}300, \geq 301$); parity_j is the fixed effect of the j th parity ($j = 1, 2, 3, \geq 4$); MY_k is the fixed effect of the k th quintile of daily MY ($k = 5.0\text{--}19.0, 19.6\text{--}23.0, 23.6\text{--}27.0, 28.0\text{--}32.0, 32.2\text{--}54.2$ kg/d); altitude_l is the fixed effect of the l th class of altitude of the herd ($l = \text{plain, hill, mountain}$); season_m is the fixed effect of the m th season ($m = \text{spring, summer, autumn, winter}$); THI_n is the fixed effect of the n th class of THI ($n = \leq 67, 68\text{--}72, 73\text{--}76, \geq 77$); ventilation_o is the fixed effect of the o th ventilation of the herd ($o = \text{ventilation, no ventilation}$); feeding system_p is the fixed effect of the p th feeding system ($p = \text{dTMR, sTMR, TRD}$); herd_q is the random effect of the q th herd ($q = 1\text{--}53$) within the l th altitude, m th season, n th THI, o th ventilation, and p th feeding system, $\sim N(0, \sigma_{\text{herd}}^2)$, where σ_{herd}^2 is the herd variance; r is the individual observation/record within the corresponding combination of fixed effects and herd; and $e_{ijklmnopqr}$ is the random residual $\sim N(0, \sigma_e^2)$, where σ_e^2 is the residual variance.

Polynomial contrasts were estimated between LSM of each mineral for the effect of DIM and MY to test linear, quadratic, and cubic trends. Trend tests evaluated ordered class effects and did not depend on the actual distance between class boundaries. Orthogonal contrasts were fitted for the effects of parity: (1) 1 vs. ≥ 2 , (2) 2 vs. ≥ 3 , (3) 3 vs. ≥ 4 ; altitude: (1) mountain vs. plain, (2) hill + mountain vs. plain; season: (1) winter + autumn vs. summer, (2) winter + autumn vs. spring, (3) winter + autumn vs. spring + summer; THI: (1) ≤ 67 vs. ≥ 68 , (2) 68 to 72 vs. ≥ 73 , (3) 73 to 76 vs. ≥ 77 ; ventilation: presence vs. absence; and feeding system: (1) dTMR + sTMR vs. TRD, (2) dTMR + TRD vs. sTMR. The P -values of the prespecified pairwise contrasts were adjusted using the Holm-Bonferroni method within each mineral and factor.

All analyses were conducted in R version 4.4.2 (R Core Team, 2024). Data management and preliminary analyses were performed using the *tidyverse* ecosystem, and linear mixed-effects models were fitted using the *lmer* function from the *lme4* package; P -values for fixed effects were obtained using the *lmerTest* package. Least squares means were computed using the *emmeans* package. Model assumptions were evaluated by visual inspection of quantile–quantile plots and residuals versus fitted plots (Supplemental Figure S1; see Notes). Residuals were approximately normally distributed, with only minor tail deviations and no evidence of severe heteroscedasticity; therefore, no transformation or specific variance structure was applied.

RESULTS AND DISCUSSION

Variability of the Composition and Mineral Profile of Milk

This study is based on an extensive sampling design encompassing 53 commercial Brown Swiss herds distributed across 3 important dairy regions of northern and central Italy, namely Emilia-Romagna, Lombardia, and Toscana. These regions represent the core production areas for some of the country's most renowned Protected Designation of Origin (PDO) cheeses, such as Parmigiano Reggiano and Grana Padano. The sampled herds covered a wide range of altitudes, from lowland plains to hilly and mountainous zones, and exhibited substantial heterogeneity in feeding systems, housing types, and management practices. This diversity mirrors the complexity of the studied production context, where Brown Swiss cows are reared under markedly different environmental and management conditions that reflect local resources and production traditions. Such a sampling strategy was adopted to capture the broad variability of the environments in which dairy cows are farmed, allowing an in-depth evaluation of whether and how environmental and herd management factors influence milk quality and mineral composition.

Descriptive statistics confirmed the expected milk composition from individual Brown Swiss cows, whereas minerals showed considerably higher variability than major constituents, reflecting both the individual variability and the diverse environmental and management conditions among herds. Specifically, the average fat content of milk was 4.13%, protein content averaged 3.70%, and casein content 2.93% (Table 2). These percentages are highly favorable for cheese production, as high fat and protein (casein) levels significantly enhance cheese yield and are related to high recovery of nutrients in the curd (Cipolat-Gotet et al., 2018; Mariani et al.,

Table 2. Descriptive statistics of milk yield, gross composition, and mineral content¹

Trait	n	Mean	SD	P5	P95
Milk yield, kg/d	1,060	25.3	7.77	13.0	38.0
Gross composition, %					
Fat	1,060	4.13	0.80	2.81	5.38
Protein	1,060	3.70	0.39	3.10	4.28
Lactose	1,060	4.74	0.24	4.34	5.06
Casein	1,060	2.93	0.33	2.40	3.46
Total solids	1,060	13.2	0.98	11.5	14.7
Mineral, mg/kg					
Ca	1,049	1,381	133	1,195	1,611
P	1,045	1,256	121	1,072	1,461
Mg	1,056	120	17.6	83.0	146
S	1,054	401	39.8	338	466
K	1,054	1,435	98.8	1,263	1,584
Cl	1,051	986	133	796	1,219
Na	1,046	421	98.0	252	600

¹P5, P95 = 5th and 95th percentiles, respectively.

2022). Selective breeding programs have increasingly prioritized cheese-making quality traits, shifting focus beyond milk yield to include milk composition, particularly casein fractions (Miglior et al., 2017). This trend reflects growing market demand for milk optimized for traditional cheese production, particularly in Europe and North America (OECD-FAO, 2025). The strong economic relevance of cheese yield, closely linked to fat and casein concentrations, has reshaped breeding objectives in countries with deep-rooted cheese-making traditions such as Italy and France. This is even more evident in the case of PDO cheese production systems. In Italy, where over 80% of milk is processed into cheese, milk composition traits are integral components of selection indices. Despite this progress, milk minerals remain largely overlooked in breeding schemes. Understanding the biological and environmental factors influencing these mineral traits could therefore provide new opportunities to refine selection programs and further enhance cheese-making efficiency.

Regarding minerals, the average concentrations observed in this study were generally higher than those reported by Carroll et al. (2006), who reported 1,170 mg/kg Ca, 300 mg/kg Na, 100 mg/kg Mg, and 1,020 mg/kg P, and by Stocco et al. (2019b), who reported 1,244 mg/kg Ca, 366 mg/kg Na, 106 mg/kg Mg, and 1,024 mg/kg P. Although the studies of Carroll et al. (2006) and Stocco et al. (2019b) were conducted in different farming systems, with distinct analytical approaches and animal populations, the differences observed in mineral concentrations could be due to a variety of factors, including potential trends over time, as well as differences in genetics and feeding practices. In addition, methodological differences should be considered when comparing mineral concentrations across studies. Indeed, previous studies often relied on inductively coupled plasma spectroscopy after acid digestion, whereas WDXRF enables direct analysis

of liquid milk, reducing preparation-related losses and matrix recovery issues. Therefore, some of the discrepancies among studies may reflect analytical methodology rather than biological variation.

The high concentrations of Ca and P (1,381 and 1,256 mg/kg, respectively; Table 2) observed in the present study are technologically favorable, as both are essential for casein micelle structure, curd formation, and coagulation efficiency, improving cheese yield and nutrient recovery (Lim et al., 2020; Christophe et al., 2021; Stocco et al., 2021). In contrast, the role of Mg (120 mg/kg) in milk processing has been less studied; although Mg contributes to ionic balance and enzymatic activity, excessive Mg concentrations have been associated with adverse effects on coagulation properties and curd firmness (Stocco et al., 2021).

Among the osmotic minerals, Na, K, and Cl displayed contrasting technological implications. Elevated Na levels are generally unfavorable, being associated with lower cheese-making efficiency (Gaucheron, 2005; Stocco et al., 2019a, 2021) and potential udder health issues (Batavani et al., 2007; Summer et al., 2009). Conversely, K plays an important role in osmotic regulation and casein stabilization, although its direct contribution to coagulation appears secondary and possibly mediated through its relationship with protein and overall mineral balance. Stocco et al. (2021) observed an apparent effect of K only after fat and casein were removed from the statistical model, suggesting that the contribution of K to coagulation and cheese-making properties is largely indirect and may instead reflect the influence of casein and the overall mineral equilibrium within milk. Chloride, though less frequently studied as an independent trait, influences milk ionic equilibrium and may affect both quality and coagulation, especially when it occurs as part of compounds such as sodium chloride (NaCl) or calcium chloride (CaCl₂; Fox et al., 2017; Stocco et al., 2019a).

Its quantification is challenging using conventional techniques such as inductively coupled plasma optical emission spectroscopy due to the element's volatility and plasma interference (Ozbek and Akman, 2016; Chen et al., 2018). In contrast, WDXRF enables direct detection of light elements such as Cl in liquid samples without digestion, providing a reliable alternative for comprehensive elemental profiling (Marguí et al., 2022). Using this approach, the average Cl concentration in the milk analyzed in this study was 986 mg/kg, providing one of the few available reference values for Brown Swiss milk under field conditions.

Individual Animal Factors

Animal-level variation is recognized as a major source of variability of milk mineral composition (Manuelian et al., 2018; Christophe et al., 2021; Toscano et al., 2023). Animal factors were incorporated into the statistical model to accurately assess the contribution of environmental and management effects while also adjusting for individual physiological differences.

The effect of DIM was significant for all the minerals analyzed ($P < 0.001$; Table 3). As lactation progressed, increasing trends were observed for Ca, P, Mg, S, and Na, whereas K showed a progressive decline (Figure 1). The LSM of milk minerals for the effect of DIM class are presented in Supplemental Table S1 (see Notes). This pattern is consistent with previous findings (Stocco et al., 2019b; Voronina, 2022) and highlights the inverse relationship between Na and K due to their complementary roles in maintaining osmotic balance across the mammary epithelium (Summer et al., 2009). The excretion pattern of Cl appeared to reflect a dilution effect, with minimum concentrations detected during the peak of lactation (Figure 1c). Although information on milk Cl dynamics is limited, Biffani et al. (2025) reported a linear increase

of this mineral throughout lactation period. However, in the same study, parity exerted a clear influence, with Cl levels increasing up to parity 3 and stabilizing thereafter.

In our study, parity showed a significant effect on P, K, Cl, and Na ($P < 0.001$; Table 3). Milk from primiparous cows contained the highest concentrations of P and K but the lowest levels of Na and Cl (Figure 2). This pattern progressively reversed in cows of fourth or greater parity. These findings align with previous studies (Manuelian et al., 2018; Stocco et al., 2019b) and support the hypothesis that age-related changes in mammary tissue, particularly the loss of tight junction integrity, lead to increased permeability of the mammary epithelium. As a result, greater influxes of Na and Cl from the extracellular fluid occur, whereas K concentrations decrease to maintain osmotic equilibrium (Armstrong, 2003). Even the inclusion of SCS and lactose as covariates did not modify the significance of the parity effect (data not shown). The effect of parity on SCS is reported in Supplemental Table S2 (see Notes).

The effect of MY was significant for all minerals ($P < 0.001$) except for P (Table 3). Concentrations of Ca, S, Mg, Na, and Cl decreased as MY increased (Figure 3), confirming the well-known dilution effect, whereby higher milk output leads to reduced mineral concentration per unit volume. In contrast, K displayed an opposite trend, increasing in proportion to milk production (Figure 3c). This may reflect the role of K in supporting osmotic regulation and secretory activity within the mammary glands of high-yielding animals, compensating for the increased water flux associated with elevated milk production.

Effects of Environmental and Herd Management Factors

Herd Effect. In most studies, the herd effect represents an important source of variability in respect to milk com-

Table 3. ANOVA (F -values and significance) of milk minerals concentration (mg/kg) for the animal (DIM, parity, and milk yield), and environmental and herd management factors (altitude, season, temperature–humidity index [THI], ventilation, and feeding system), along with the percentage of total milk mineral variance accounted for by the herd effect

Mineral	Animal factor			Herd ¹ (% of total variance)	Environmental and herd management factor					RMSE ²
	DIM	Parity	Milk yield		Altitude	Season	THI	Ventilation	Feeding system	
Ca	5.95***	0.88	6.14***	0.01	0.45	4.57**	1.01	0.02	1.15	113
P	6.05***	12.9***	1.74	0.00	1.01	5.56**	0.80	3.28†	5.94**	107
Mg	20.0***	1.33	8.01***	0.02	0.38	3.92*	6.90***	0.70	3.20†	12
S	51.3***	0.20	15.2***	0.01	0.77	6.98***	0.47	1.64	6.33**	29
K	27.7***	46.7***	12.7***	3.88	2.60†	7.89***	1.25	3.72†	1.97	74
Cl	4.95***	92.8***	9.68***	0.01	0.99	0.33	3.41*	0.17	0.46	104
Na	14.8***	57.7 ***	17.8***	1.30	0.93	0.04	0.82	0.16	3.05†	77

¹Herd (% of total variance) indicates the percentage of the total variance attributed to the herd random effect.

²RMSE = root mean square error.

† $P < 0.10$; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

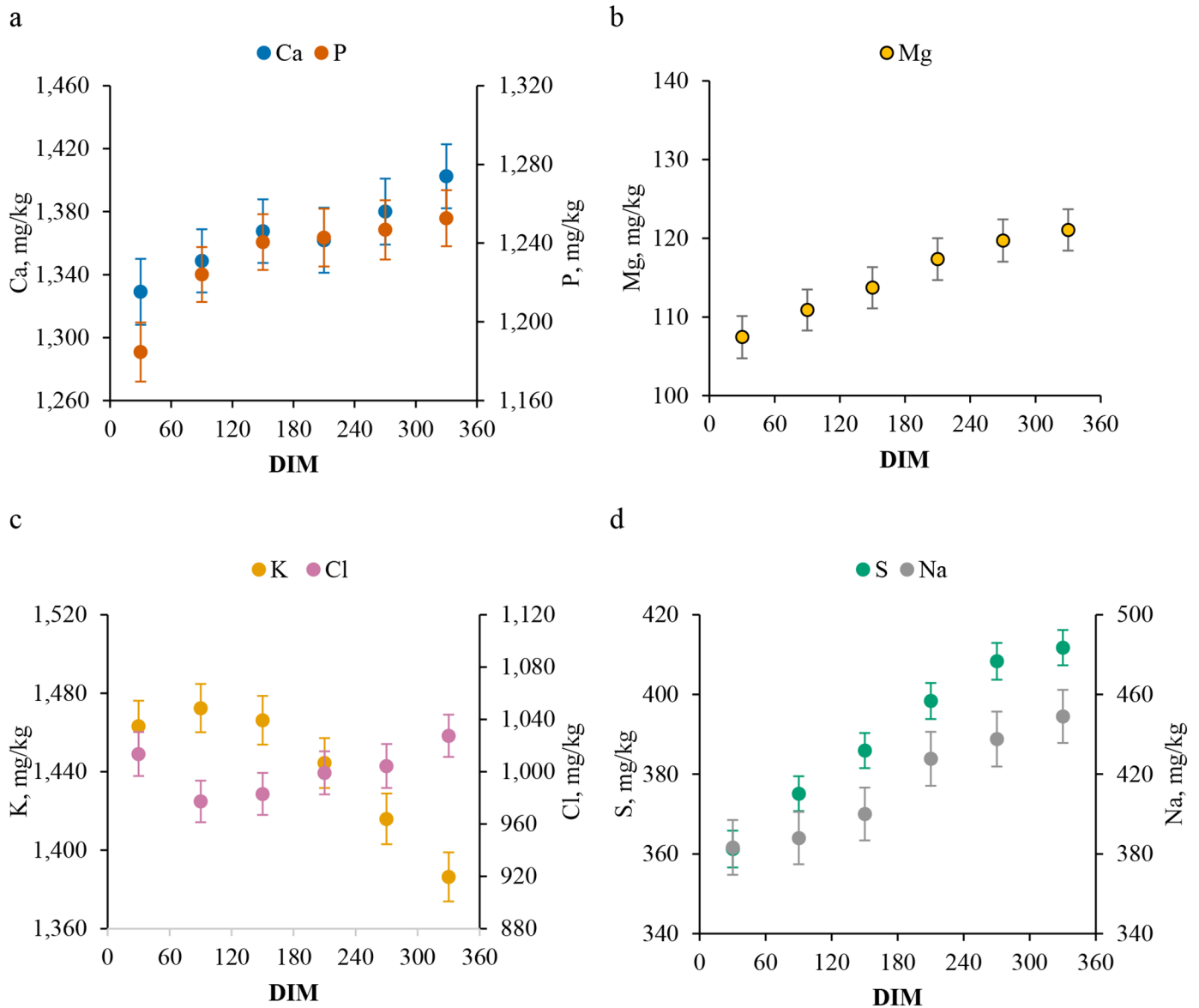


Figure 1. Least squares means (with SE bars) of the concentration of milk (a) Ca and P, (b) Mg, (c) K and Cl, and (d) S and Na for the effect of DIM.

position, as it encompasses a wide range of environmental, nutritional, and management conditions that collectively influence milk minerals. Each herd can be viewed as a distinct production ecosystem, defined by its specific combination of feeding strategy, housing type, climate, and herd management practices. In the present study, the inclusion of more than 50 herds distributed across different production areas allowed for a comprehensive representation of the variability that characterizes Italian dairy systems. Thanks to the detailed herd-level information collected through structured surveys, it was possible to model many of the underlying environmental and management factors individually, such as altitude, season,

THI, ventilation, and feeding system, rather than treating them as a single composite herd effect. As a result, the variance attributed to the herd random effect accounted for only 0.00% to 3.88% of the total variability (Table 3), considerably lower than in previous reports where herd was modeled as a broader random term (van Hulzen et al., 2009; Stocco et al., 2019b; Toscano et al., 2023). This result should be interpreted as a consequence of model specification: by partitioning herd-related heterogeneity into identifiable covariates, the residual between-herd variance reflects only unexplained variability rather than the full set of environmental differences among herds. This reduction indicates that much of the variabil-

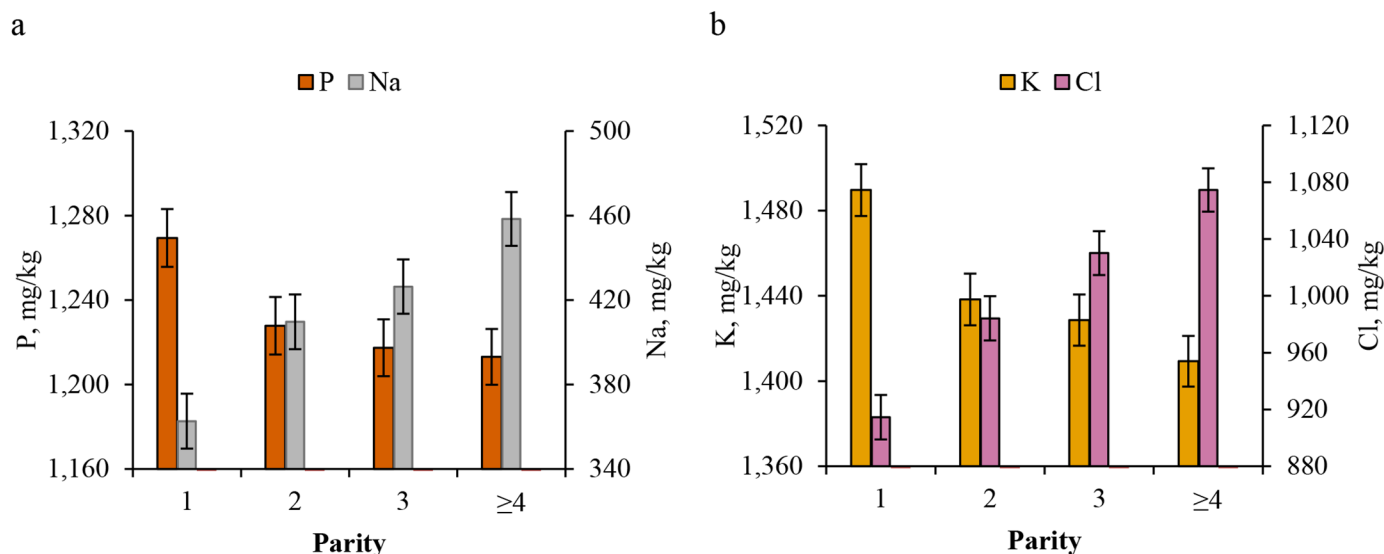


Figure 2. Least squares means (with SE bars) of the concentration of milk (a) P and Na, and (b) K and Cl for the effect of parity.

ity typically assigned to the herd level was successfully partitioned among specific, measurable factors, thereby providing a clearer understanding of how individual environmental and management components contribute to milk mineral composition.

Altitude Effect. As regards the environment, altitude appeared to be associated with the composition of milk, with differences observed in K concentration ($P < 0.10$; Table 3). The pairwise contrasts reported in Table 4 and Figure 4a showed higher K concentrations when comparing hill and mountain versus plain (+28.6 mg/kg, $P < 0.05$), indicating an association between altitude and K concentration in milk. Potassium, the predominant intracellular cation in milk, is essential for osmotic balance, mammary epithelial transport, and udder health, and its secretion into milk is precisely regulated by metabolic and environmental cues. The mechanism explaining the observed increase in K concentration with altitude remains uncertain, but several explanations are plausible. Although mountain farming systems typically rely on roughage-based diets, mainly hay, previous studies have shown that hay from intensively managed lowland grasslands can actually contain higher K concentrations than alpine hay (29 vs. 12 g/kg DM; Collomb et al., 2008; Leiber et al., 2009), suggesting that dietary intake alone cannot account for the trend observed. It is therefore likely that climatic and physiological factors contribute to the pattern. Cooler temperatures and lower oxygen partial pressure at higher elevations may influence endocrine and renal regulation of electrolytes, modifying systemic and mammary ion transport mechanisms (Ramirez et al., 1992; Gulick et al., 2016). These adjustments could pro-

mote increased K retention or secretion into milk as part of the cow's adaptive response to environmental stress.

Because studies investigating altitude-related effects on milk minerals are scarce, further research integrating forage composition, cow metabolism, and climatic data is needed to clarify the underlying biological mechanisms.

Season Effect. Season exerted a clear and multifaceted influence on milk mineral composition, significantly affecting Ca, P, Mg, S, and K ($P < 0.05$; Table 3). Concentrations of Ca and P were highest in autumn and lowest in spring and summer (Figure 4b). In particular, autumn and winter showed higher Ca and P than spring (+62.4 and +43.9 mg/kg, respectively; $P < 0.001$). These findings are consistent with those reported by Poulsen et al. (2015), who demonstrated that seasonal fluctuations in Ca and P are closely linked to changes in casein fractions, which typically decrease during warmer months and rise again in cooler periods. In contrast, Mg concentrations peaked in winter and declined in summer (Figure 4c). Magnesium concentrations were higher when autumn and winter were compared together against the combined spring and summer period (+9.76 mg/kg; $P < 0.01$). Because fresh-pasture feeding was minimal in the herds included in this study, these seasonal differences are likely related to environmental and physiological factors rather than diet composition. Climatic conditions during the summer months can impair Mg absorption and metabolism by reducing feed intake, altering gastrointestinal function, and increasing electrolyte loss through sweat and urine (Tao and Dahl, 2013). These disruptions in mineral homeostasis reduce the Mg available for milk secretion. Conversely, winter temperatures are associated with

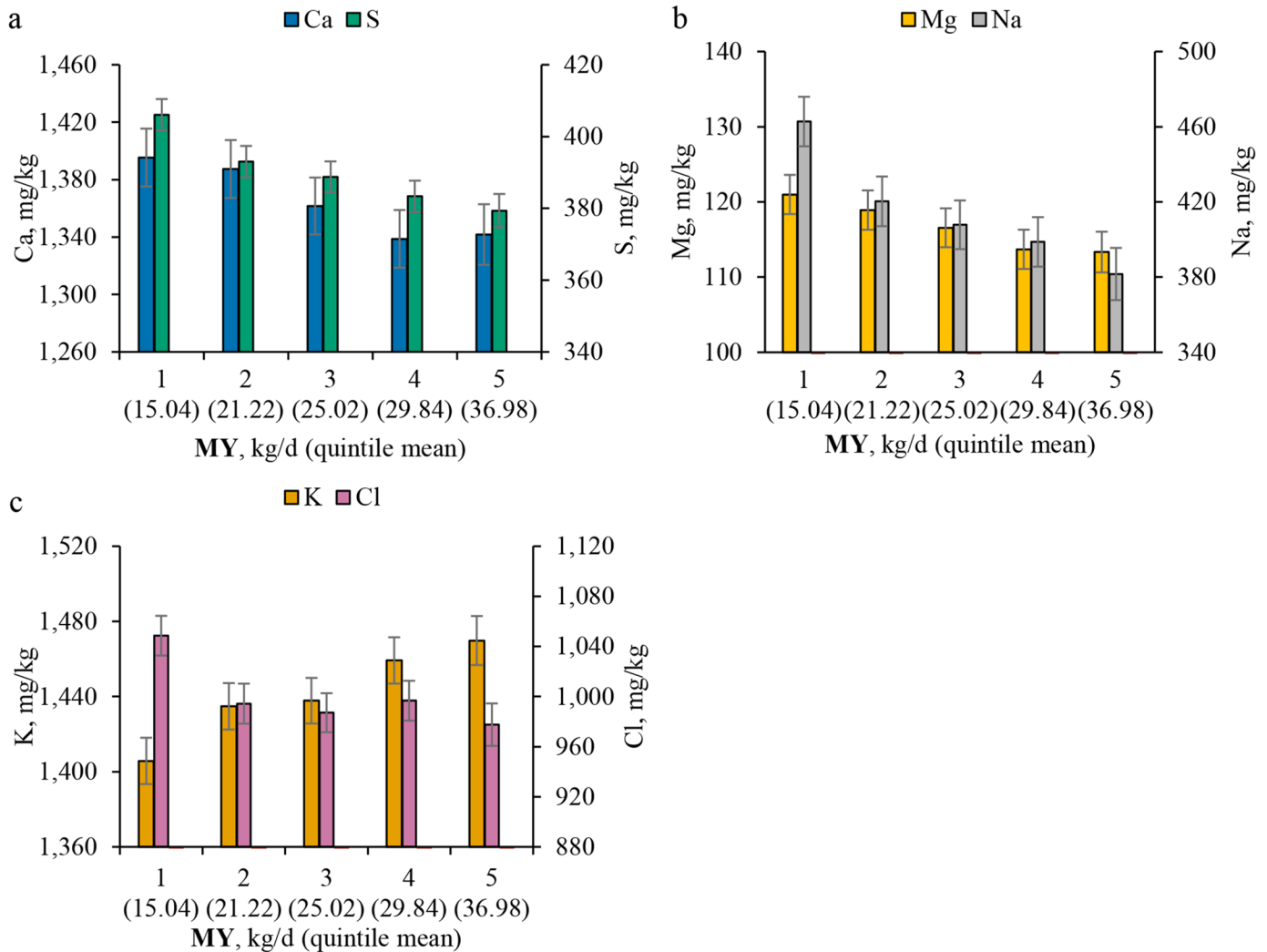


Figure 3. Least squares means (with SE bars) of the concentration of milk (a) Ca and S, (b) Mg and Na, and K and Cl for the effect of daily milk yield (MY).

more stable metabolic conditions and improved mineral utilization, leading to higher Mg levels in milk.

Concentrations of both K and S increased progressively from spring to winter, although their seasonal peaks occurred at different times: S reached its maximum in autumn and K in winter (Figure 4d). Compared with spring, autumn and winter showed higher S and K (+13.8 and +54.2 mg/kg, respectively; $P < 0.001$). These patterns likely reflect combined environmental effects and adaptive physiological responses. Elevated K during colder months may result from reduced thermal stress, leading to more stable electrolyte regulation and enhanced mammary K secretion (Moreno-Rojas et al., 1994; Harrison et al., 2011). In addition, the conserved forages typically used in winter diets ensure a consistent mineral supply, supporting this trend. The autumnal rise in S may relate to shifts in rumen microbial protein

synthesis and sulfur-containing amino acid metabolism, which affect the partitioning of S into milk (Gaucheron, 2005; Jo et al., 2019). The subsequent decline in S during winter, accompanied by rising K levels, suggests distinct regulatory mechanisms for these minerals, reflecting their different metabolic roles and excretion pathways. Overall, the strong seasonal influence observed in this study highlights the dynamic interaction between environmental conditions and the cow's physiological adaptation, further emphasizing that mineral composition in milk is not static but responds to cyclical environmental stressors and management-related changes throughout the production year. Seasonal contrasts were consistent with what emerged based on ANOVA. The comparisons “autumn and winter vs. spring” and “autumn and winter vs. spring and summer” indicated marked increases in Ca and P, as well as higher Mg and S and K levels, confirm-

Table 4. Results of contrasts for mineral concentrations (mg/kg), including estimated values for each mineral across specific contrasts of animal factors (DIM, parity, and milk yield), and environmental and herd management factors (altitude, season, temperature–humidity index [THI], ventilation, and feeding system)

Tested factors and contrast estimates	Ca	P	Mg	S	K	Cl	Na
DIM							
Linear	455***	410***	98.0***	365***	−575***	167	505***
Quadratic	11.2	−218	−12.2	−55.8	−283	295***	24.7
Cubic	171	172	−7.04	−29.5	98.1	−184	−130
Parity							
1 vs. ≥2	10.1	50.0***	1.08	0.48	64.3***	−115***	−68.8***
2 vs. ≥3	7.31	12.6	−0.61	−1.58	19.3	−68.2	−32.7
3 vs. 4	9.32	4.31	−1.80	0.83	19.3	−44.5	−32.0
Milk yield							
Linear	−156***	−61.1*	−20.5***	−63.1***	152.5***	−139.7***	−183.8***
Quadratic	24.8	2.19	2.89	17.0	−18.5	87.1	54.1
Cubic	43.7	40.0	2.78	−7.39	15.2	−75.8	−37.9
Altitude							
Mountain vs. plain	−19.0	−23.7	−3.17	−6.72	33.1	7.15	−22.6
Hill + mountain vs. plain	−4.43	−11.6	−1.99	−5.27	28.6†	16.8	−17.9
Season							
Autumn + winter vs. summer	25.3	50.4	15.1	11.6	25.1	2.48	−1.16
Autumn + winter vs. spring	62.4***	43.9***	4.39	13.8***	54.2***	13.0	0.50
Winter + autumn vs. spring + summer	43.9	47.2	9.76**	12.7	39.7	7.74	−0.33
THI							
≤67 vs. ≥68	18.1	−11.0	0.04	5.44	−15.6	−21.2	9.79
68–72 vs. ≥73	48.2	−30.2	19.7***	5.65	−15.3	−75.5*	4.06
73–76 vs. ≥77	−11.4	0.76	3.77	8.36	23.6	−59.9	−28.4
Ventilation							
Presence vs. absence	3.44	33.1†	3.15	7.64	33.1†	8.81	−7.43
Feeding system							
TMR vs. traditional	5.74	1.36	−2.15	1.44	−23.6	16.1	35.5*
Dry TMR + traditional vs. silage TMR	20.9	36.6*	7.08*	11.5*	28.2	−8.65	−20.4

† $P < 0.10$; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

ing that cooler seasons are associated with richer mineral profiles (Table 4).

Temperature–Humidity Index Effect. The influence of THI on milk mineral concentrations was significant for Mg and Cl ($P < 0.05$; Figure 5a). The LSM of Mg and Cl along with their 95% CI are presented in Supplemental Figures S2 and S3 (see Notes). The THI classes included 700 (≤67), 100 (68–72), 140 (73–76), and 120 (≥77) observations (Table 1). Notably, the THI range of 68 to 72 represents an important threshold beyond which substantial differences in milk mineral concentrations occur. This THI range corresponds to the onset of heat stress in dairy cattle (Armstrong, 1994) and can cause changes in mineral metabolism as the cows initiate physiological adaptations. Elevated Mg levels observed within this THI range likely reflect a compensatory physiological response aimed at maintaining electrolyte and metabolic homeostasis under moderate summer conditions, characterized by high ambient temperatures, elevated humidity, and intense solar radiation (Khiaosa-Ard et al., 2023). In contrast, the concurrent decline in Cl concentration at this threshold may be linked to alterations in acid–base equilibrium or shifts in mammary epithelial ion transport mechanisms as cows adjust to the rising thermal load. As heat stress intensifies beyond this threshold, reduc-

tions in feed intake, modifications in mineral absorption, and changes in systemic electrolyte regulation become more pronounced. In such conditions, decreased ruminal efficiency may reduce Mg bioavailability or increase its excretion, resulting in a decline in milk Mg concentration. Conversely, Cl concentrations tend to rise at higher THI values, possibly reflecting enhanced Cl mobilization or modified osmotic regulation aimed at maintaining plasma and mammary homeostasis during more severe thermal stress. These opposing responses of Mg and Cl across increasing THI classes highlight a complex, mineral-specific adaptation to thermal conditions. Interestingly, Joo et al. (2021) reported a decrease in blood Cl concentrations under high-temperature conditions in both Holstein and Jersey cows. Although this may appear contradictory, it likely reflects compartment-specific regulation, where Cl is preferentially directed toward milk secretion as part of mammary osmotic adjustment, and systemic levels decline through redistribution or renal compensation. Overall, these findings underscore the complexity of mineral homeostasis under varying thermal environments.

Ventilation Effect. The presence of ventilation showed a trend toward a marginal association with K and P concentrations in milk ($P < 0.10$; Table 3), with

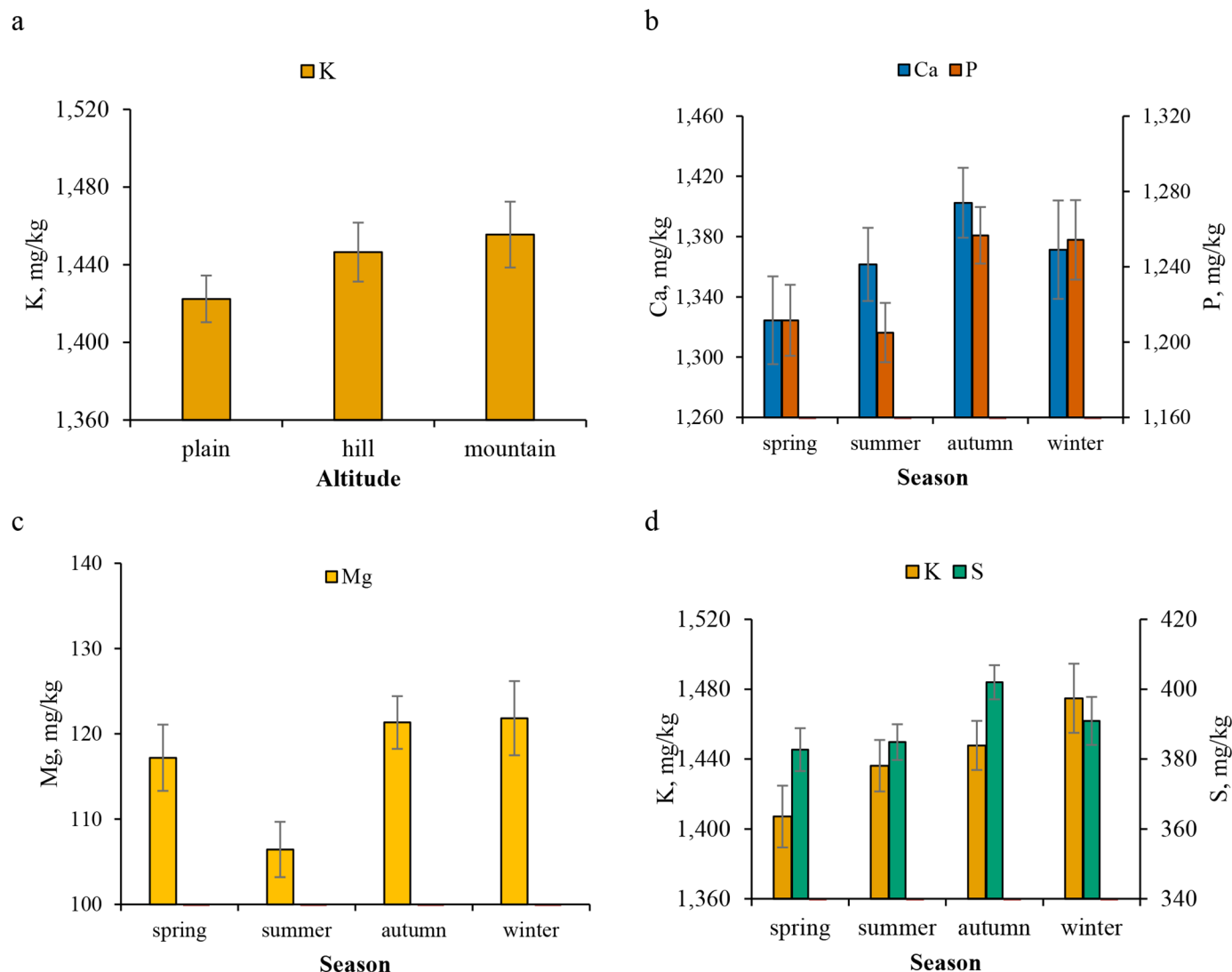


Figure 4. Least squares means (with SE bars) of the concentration of milk (a) K for the effect of altitude, and (b) Ca and P, (c) Mg, and (d) S and K for the effect of season.

higher levels observed in cows reared in ventilated herds (+33.1 mg/kg for both minerals; Figure 5b). This finding agrees with the well-established understanding that environmental management practices improving animal comfort can positively influence milk composition (Armstrong, 1994; Collier et al., 2006). The underlying mechanisms could be related to improved thermal comfort and physiological status of cows maintained in well-ventilated housing, where heat stress is mitigated. By enhancing air movement and moderating barn microclimate, ventilation supports normal feeding behavior and nutrient digestion, which may indirectly influence nutrient availability for milk synthesis and mineral transfer into milk (West, 2003; Wheelock et al., 2010).

Given the close biological links between minerals such as K and P and milk productivity, particularly the role of P in energy metabolism and protein synthesis, their higher concentrations under ventilated conditions may reflect improved metabolic efficiency and mammary function. Beyond their physiological significance, these changes also carry technological implications. Elevated K concentrations have been associated with a reduction in lactose recovery during curd formation (Stocco et al., 2025), an effect that can be advantageous in the production of hard, low-moisture cheeses, such as those typical of Italian PDO systems.

Feeding System Effect. The aim of this study was not to evaluate the effect of specific nutrient compositions but rather to assess the variability in milk mineral content

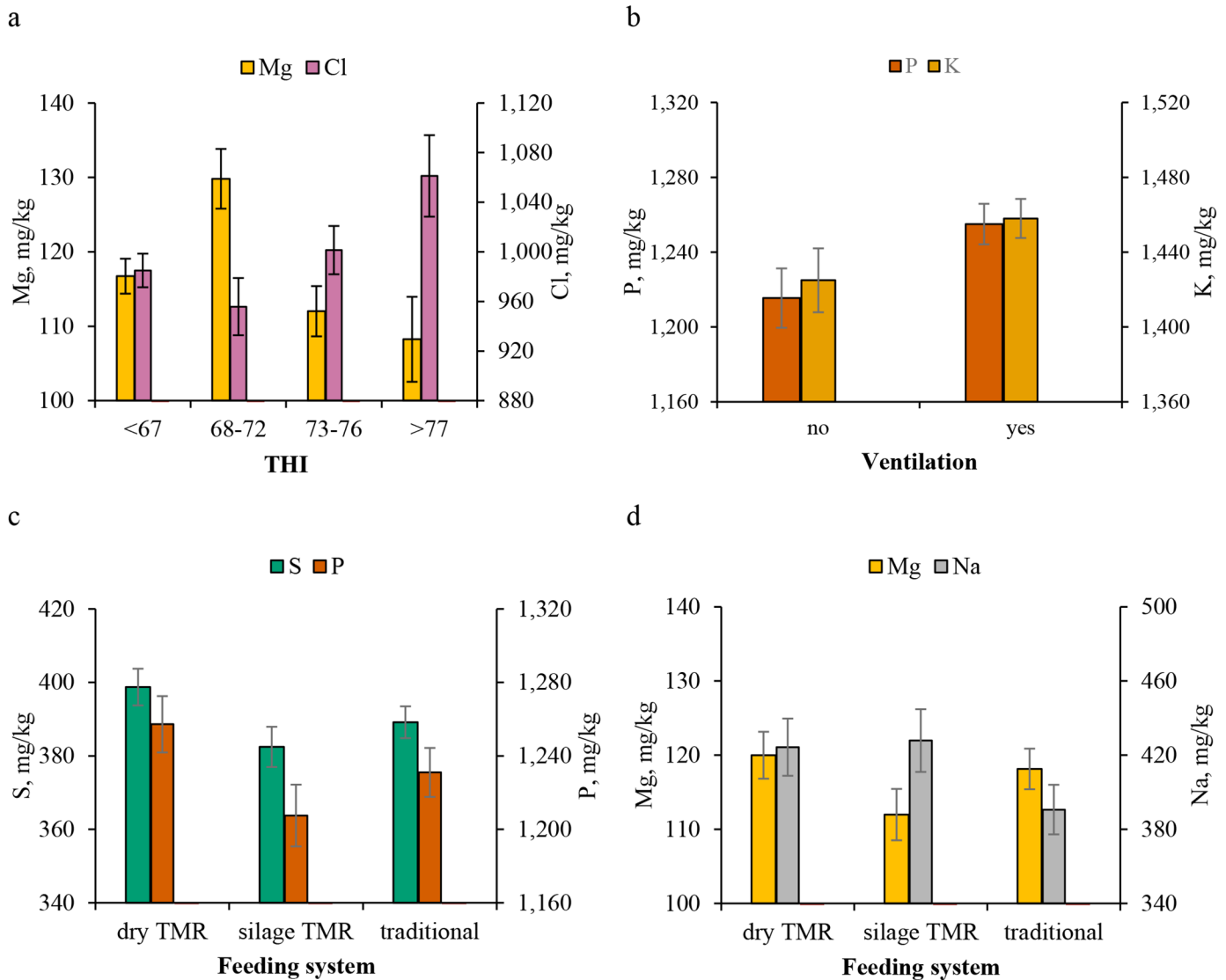


Figure 5. Least squares means (with SE bars) of the concentration of milk (a) Mg and Cl for the effect of temperature–humidity index (THI), (b) P and K for the effect of ventilation, and (c) S and P, and (d) Mg and Na for the effect of feeding system.

associated with the broad feeding systems commonly adopted in Italian dairy production. Although detailed feed composition was not recorded, this approach allowed meaningful interpretation of how macro-level feeding strategies influence milk mineral variability.

Feeding system significantly affected P and S ($P < 0.01$), with a trend toward significance for Mg and Na ($P < 0.10$; Table 3). Higher concentrations of P, S, and Mg were found in milk from cows fed dTMR and TRD feeding systems compared with those fed sTMR (+36.6, +11.5, and +7.08 mg/kg, respectively; Table 4, Figure 5c). This finding aligns with previous reports showing that dried forages generally have higher mineral concentrations (on a DM basis) than maize silage (Burke et al., 2007; Manzocchi et al., 2020).

Milk from cows on TRD feeding system, characterized by ad libitum access to roughages and separate concentrate feeding, exhibited particularly high levels of P, S, and Mg compared with sTMR. Differences in mineral excretion between feeding systems are well documented: sTMR often result in lower milk mineral concentrations, likely due to reduced mineral bioavailability or altered rumen metabolism (De La Torre-Santos et al., 2021). Conversely, milk from cows fed dTMR displayed the highest mineral content overall, possibly reflecting the lower likelihood of feed-sorting behavior and more consistent nutrient intake due to the physical characteristics of the ration (Manzocchi et al., 2020).

Sodium concentration was significantly higher in milk from cows fed TMR, regardless of whether the TMR was

based on dry or ensiled forages (+35.5 mg/kg; Table 4, Figure 5d). Typically, TMR systems, particularly those low in fresh forage, exhibit a lower K:Na ratio, which may alter mammary epithelial ion transport and promote increased Na secretion into milk (Gulati et al., 2018). Alternatively, the higher Na concentration could partly reflect the inclusion of salt or mineral supplements commonly used in TMR formulations, although ration composition was not directly quantified in the present study. Feeding strategy contrasts showed clear differences, especially in the case of Na when comparing TMR versus TRD and for P, Mg and S from the comparison of dTMR and TRD versus sTMR. Specifically, the comparison dTMR and TRD versus sTMR indicated higher P, Mg, and S, supporting the hypothesis that diets based on dry forage or traditional feeding systems result in a richer mineral profile than silage-based TMR (Table 4).

Overall, these results indicate that, although TMR systems are economically advantageous and support higher yields, they may adversely affect milk mineral balance and, consequently, its nutritional and technological quality. This finding highlights a trade-off between production efficiency and milk compositional integrity (Stone, 2003; Bach, 2024).

CONCLUSIONS

This study comprehensively evaluated the combined influence of animal, environmental, and management factors on the mineral composition of milk from Brown Swiss cows, specifically within production environments representative of Italian PDO cheese systems. By integrating detailed herd-level data with individual milk analyses, the study provided a holistic view of how physiological and external factors interact to modify mineral profiles. Although animal factors (e.g., parity, milk yield) explained significant variability, environmental and management conditions exerted quantifiable, mineral-specific influences. Season affected most minerals (Ca, P, Mg, S, and K), followed by feeding system (P and S) and THI (Mg and Cl). Ventilation showed a marginal association with K levels. Because these variables were explicitly modeled, the low residual herd variance indicates that measurable conditions explain most between-herd differences. These findings underscore the importance of considering environmental factors when interpreting milk quality. Future research integrating genetic data will further clarify these interactions, supporting sustainable breeding and precision management strategies.

NOTES

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Food and Forestry Policies (MiPAAF; Rome, Italy), under project code D94119000840001. Supplemental material for this article is available at <https://doi.org/10.6084/m9.figshare.32160390>. The cows involved in this study were not subjected to any invasive procedures, and milk samples were collected noninvasively by trained farm technicians as part of routine milking practices. No additional handling or intervention beyond standard farm operations was required, and thus approval by an Institutional Animal Care and Use Committee or Institutional Review Board was not required. The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: a.s.l. = above sea level; dTMR = dry TMR; MY = milk yield; PDO = Protected Designation of Origin; RMSE = root mean square error; sTMR = silage TMR; THI = temperature–humidity index; TRD = traditional feeding system; WDXRF = wavelength-dispersive X-ray fluorescence.

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